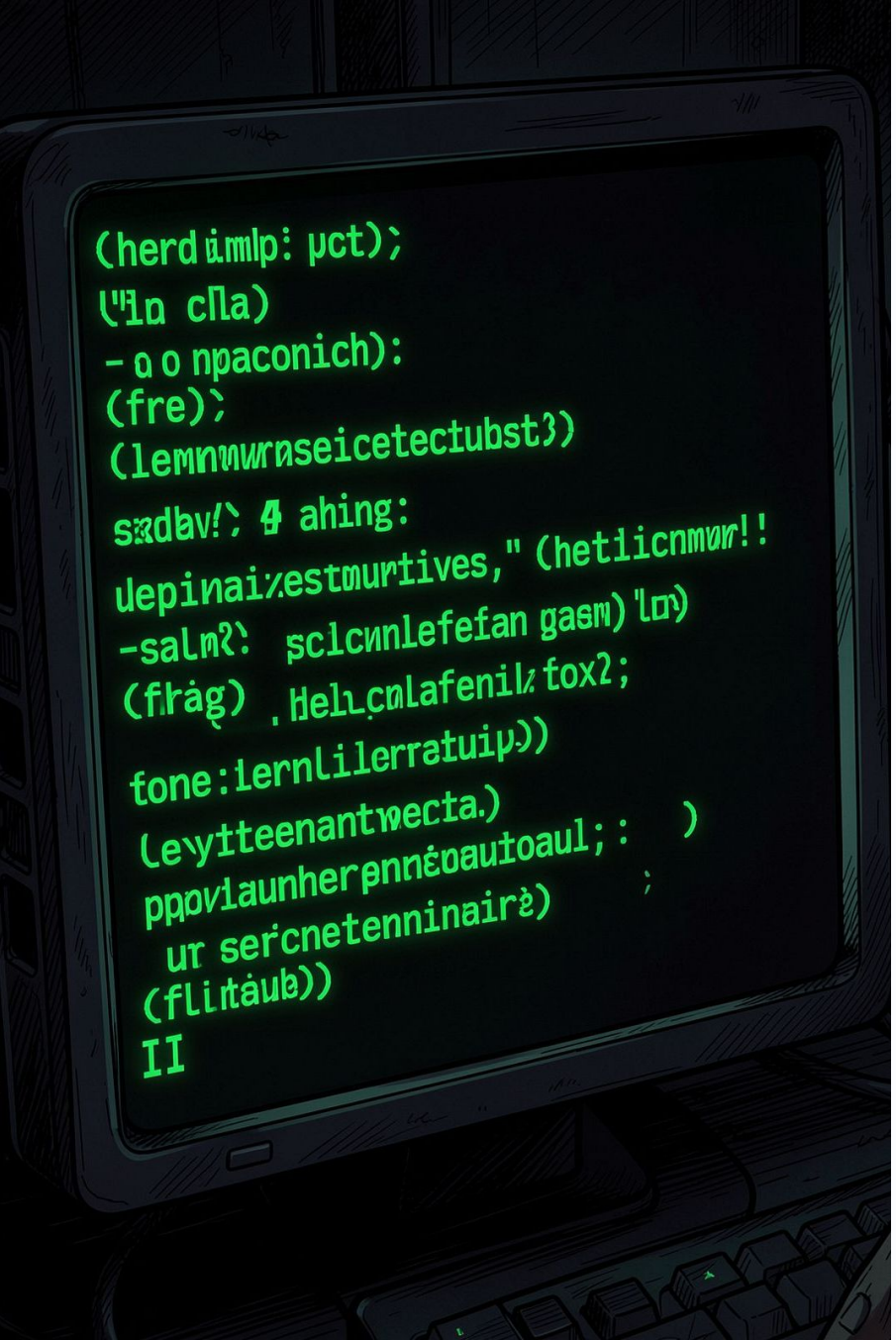


Linux Command-Line Mastery

Essential Navigation, File Management, and System Inspection

2-HOUR INTERACTIVE WORKSHOP | 28 COMMANDS | 5 PHASES

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PHASE 1 · NAVIGATION

Where Am I, and How Do I Move?

`pwd` — Print Working Directory

Shows your current path. Use `pwd -P` to resolve symbolic links.

`ls` — List Directory Contents

`ls -l` detailed · `ls -a` hidden files · `ls -h` human sizes

Pro tip: `ls -lah` — your daily driver.

`cd` — Change Directory

`cd /home/user` absolute · `cd ~` home · `cd ..` up one level

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PHASE 1 · FILE MANAGEMENT

Creating and Organizing Your Data



`mkdir` — Make Directories

`mkdir project` creates one folder. Use `mkdir -p parent/child` to build nested paths in one step.

`touch` — Create Empty Files

`touch notes.txt` creates an empty file or updates its timestamp if it already exists.

`cp` — Copy Files and Folders

`cp file.txt backup.txt` copies a file. Add `-r` or `-R` to copy directories recursively.

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PHASE 1 · SAFETY FIRST

Moving, Renaming, and Deleting

`mv` — Move and Rename

Renames or relocates files and directories.

```
mv oldname.txt newname.txt
mv file.txt /home/user/docs/
```

Use `mv -i` for interactive confirmation before overwriting.

`rm` — Remove Files and Directories

```
rm file.txt
rm -r folder/
rm -rf folder/
```

⚠ ⚠️ **Critical:** `rm -rf` deletes without confirmation. Always double-check your path. There is no recycle bin in the terminal.

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PHASE 2 · FILE INSPECTION

Reading and Inspecting File Contents

1

`cat`

Prints full file contents. `cat -n file.txt` adds line numbers.

2

`less`

Paginated viewer for large files. Scroll with arrow keys, quit with `q`.

3

`head` / `tail`

First/last 10 lines by default. `tail -f logfile.log` streams live updates.

4

`wc`

Word count. `wc -l` lines, `wc -w` words, `wc -c` characters.

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PHASE 3 · SEARCH AND FILTER

Finding Files and Text Patterns

`grep` — Search Text Patterns

```
grep "error" logfile.txt
grep -i "warning" *.log
grep -rn "TODO" /project/
```

- `-i` — case-insensitive
- `-r` — recursive through directories
- `-n` — show line numbers

`find` — Locate Files by Name or Type

```
find /home -name "*.txt"
find . -type f
find . -type d -name "logs"
```

- `-name` — match filename patterns
- `-type f` — regular files only
- `-type d` — directories only

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PHASE 4 · SYSTEM INFO

Checking Host Status and Disk Usage



`uname -a`

Displays kernel version, hostname, architecture, and OS details in a single line.



`df -h`

Shows disk space on all mounted filesystems in human-readable format (GB, MB).



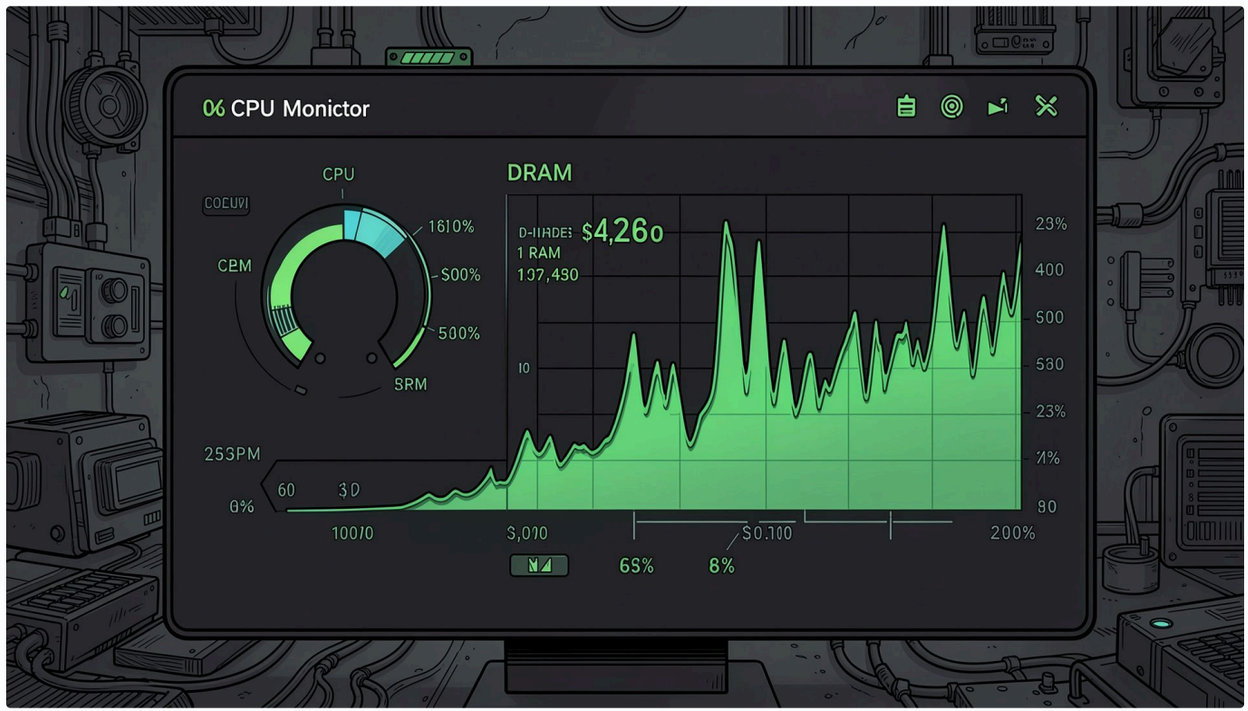
`du -sh`

Summarizes total disk usage of a directory. `du -sh /var/log` checks a specific folder.

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PHASE 4 · PROCESS MONITORING

Managing System Performance



free -h

Displays RAM and swap usage in human-readable format.

top / htop

Real-time interactive process viewer. `htop` offers a friendlier interface.

ps aux

Snapshots all running processes with CPU, memory, and PID info.

kill

Terminates a process by PID. `kill -9 PID` force-kills unresponsive processes.

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PHASE 5 · SECURITY AND NETWORKING

Permissions, Ownership, and Connectivity

chmod

Change file access modes. `chmod +x script.sh` makes a file executable. Use `-R` for recursive changes.

chown

Change file owner and group. `chown user:group file.txt` transfers ownership.

ping

Test network connectivity. `ping -c 4 google.com` sends exactly 4 packets.

sudo

Execute commands with elevated (root) privileges. Required for system-level changes.

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WORKSHOP WRAP-UP

Key Takeaways and Golden Rules

28 Commands Covered

Navigation · File Management · Inspection · Search · System Info · Processes · Permissions

Golden Rules

- Read error messages — they tell you what went wrong
- Use `-i` flags for interactive safety on `mv` and `rm`
- Always verify your path before running destructive commands
- Use `--help` or `man` when unsure

🎓 **Open Q&A** — Now it's your turn. What commands will you practice first?

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